

Review

Review on the Application of Living PSA in Nuclear Power

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Abstract: With the increasing standards of safety management in nuclear power plants, Living Probabilistic Safety Assessment (Living PSA) technology has begun to play an increasingly important role in their operation. This paper aims to provide an overview of the application and development of Living Probabilistic Safety Assessment (Living PSA) technology in nuclear power plant safety monitoring and risk assessment, examining the key technologies and future challenges. Initially, we summarize the current safety needs in regard to nuclear power, examine the policy on configuration risk management technology for nuclear power plants, and outline its importance and development process in nuclear power plant safety management. Subsequently, we discuss the basic principle of Living PSAs and the working method of risk monitoring based on Living PSAs, including information monitoring data collection, online identification, real-time model updating, and risk calculation. Within the Living PSA framework, model development is not merely about creating a theoretical or static representation; it is a dynamic and ongoing process that involves a deep understanding and precise simulation of the behavior of nuclear power plant systems and components. This represents the main research efforts in Living PSAs at present. Additionally, this paper identifies the key technologies of Living PSAs in an in-depth manner, such as the reliability-model-updating technology and model building in dynamic reliability analyses, including the fault tree model, multi-layer flow model, GO-FLOW model etc. The paper lists the work of some scholars in this area in recent years, which helps readers and researchers to clearly understand the current progress of Living PSA technologies in terms of model establishment and updating. Finally, the paper summarizes the challenges and future development of Living PSA and emphasizes the possible problems in data quality, human factor engineering, and the development of Living PSA technologies in the future. In the future, Living PSAs will provide more solid support for the realization of safer and more economical methods of operating nuclear power plants.

Keywords: living PSA; nuclear safety; reliability analysis



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1. Introduction

1.1. Nuclear Power Development and Nuclear Safety

The establishment of the first controlled nuclear chain reaction reactor, Chicago Pile-1, in 1942 marked the beginning of the nuclear energy era for humanity [1]. Compared to other energy sources, nuclear energy offers advantages, such as being clean, efficient, and stable, yet it also faces challenges, such as high initial costs for plant construction, complex waste management issues, and potential safety concerns in the event of accidents [2]. As of the end of 2023, 413 nuclear power units are operational worldwide, with a total installed capacity of 371.51 million kilowatts, distributed across 31 countries [3]. Currently, widely used reactor technologies include pressurized water reactors, boiling water reactors, and heavy-water reactors, with pressurized water reactors being the most commonly utilized [4].

As of July 2024, there are 59 nuclear power units under construction worldwide, mainly in China and India, as well as in Russia [5]. According to the “World Nuclear Industry Status Report 2020”, 270 reactors have been in operation for 31 years or longer [3,6]. Although the nuclear energy industry offers various benefits, its potential safety issues cannot be overlooked. Hence, from the very inception of the nuclear industry, nuclear power has faced significant opposition and skepticism. As the number of nuclear facilities continues to grow, and with older nuclear plants still being in operation, the development and implementation of advanced safety technologies have become increasingly crucial. This focus is a major area of work for many contemporary nuclear industry professionals and scholars.

Following the Fukushima nuclear incident, newer and higher standards for nuclear safety have been established by the international community. According to the Generation IV International Forum, the Generation IV nuclear energy systems are designed to achieve the goals of sustainability, economic viability, safety, reliability, non-proliferation, and physical protection [7]. Currently, the application of nuclear power plants is moving towards greater safety, economic efficiency, and the development of small modular reactors [8,9]. As the number of nuclear power plants increases and the lifespan of existing ones is extended, ensuring safety becomes even more critical. In the field of nuclear safety, real-time risk monitoring and configuration risk management technologies have become key to solving the comprehensive risk assessment and management challenges in nuclear power plants. This technology integrates sensing, big data, artificial intelligence, and modeling techniques to enable real-time risk identification and quantification. By dynamically adjusting management strategies, nuclear power plant risks are ensured to be “knowable and controllable”, thereby providing technical support for the safe utilization of nuclear energy.

1.2. Background of Living PSA

Currently, the risk-based configuration risk management technologies implemented in the nuclear industry operate based on Living PSA models [10]. According to IAEA-TECDOC-1106 [11,12], a Living PSA is a dynamically updated safety assessment method that reflects the current design and operating characteristics of nuclear power plants. The implementation of a Living PSA needs to be documented so that each part of the model can be correlated with actual data from the plant or information related to the assumptions of analysts when data are insufficient. According to the policy on configuration risk management technology for nuclear power plants (for Trial Implementation) [12,13], “living PSA: To fully reflect the current situation of nuclear power plants, the probabilistic safety analysis method is applied during the operation of nuclear power plants, using new technical information, more accurate methods and tools, combined with changes in the design and operation of nuclear power plants, and the new information obtained to update the probabilistic safety analysis model and data evaluation methods”. Configuration risk management technology involves a comprehensive risk assessment and management method. It identifies and manages potential risks by analyzing and monitoring changes in the configuration of nuclear power plants to ensure their safe operation.

Living PSAs have evolved from probabilistic safety assessments (PSAs); they are not a new concept but an extension of traditional PSAs. Compared to a traditional PSA, a Living PSA features real-time data updates and the ability to respond to design and operational changes. A comparison of the characteristics of a traditional PSA and a Living PSA is shown in Table 1.

Compared with a traditional PSA, a Living PSA is prioritized by nuclear power plants to address changes in plant system design and operating procedures, scope improvements for PSA analyses, as well as variations in system equipment statuses.

Table 1. Comparison of the characteristics of traditional PSA and Living PSA [11,12].

Character	Traditional PSA	Living PSA
Timeliness	Evaluating during the design and construction phases or periodically, with infrequent updates	Continuously conducted with real-time updates to models and data
Flexibility	Response is slow, primarily used for long-term planning and strategic decision-making	Highly flexible and capable of rapidly responding to changes in design and operation
Source and processing of data	Relies on pre-collected and organized data, with relatively static processing and a low update frequency	Integrates real-time sensor data, big data analytics, and artificial intelligence technologies to dynamically process data
applied range	Safety assessments during the design phase, permit applications, and long-term risk management strategies	Safety assessment during the design phase, permit application, and long-term risk management strategies

1.3. Development of Implementing Living PSA

In the 1950s, the nuclear engineering education in the U.S. started to make the quantitative measurement of reactor risk and reliability a priority and began to focus on reactor reliability [14]. Currently, Living PSAs are now widely used in nuclear power plants. However, the aerospace industry was the first to apply probabilistic methods for safety assurance. In the 1960s, Boeing and Bell Labs pioneered the use of fault tree analyses, which were later applied in the manufacturing process of the Boeing 747. After the Apollo 1 accident in 1967, NASA (National Aeronautics and Space Administration) began to seek out methods of probabilistic risk assessment; however, discrepancies between results and actual experience led to skepticism about probabilistic risk assessments. From 1972 to 1975, under the leadership of Professor N. C. Rasmussen at MIT (Massachusetts Institute of Technology), fault tree probabilistic safety analyses were conducted for the Surry and Peach Bottom nuclear power plants in the U.S., culminating in the publication of the WASH-1400 report, which did not receive significant attention [15].

The Three Mile Island accident (1979) and the Chernobyl disaster (1986) highlighted the value of the WASH-1400 report. These incidents also prompted U.S. nuclear power plants to adopt probabilistic analysis, significantly advancing the development of PSA in the nuclear industry [16]. In 1986, the NRC (Nuclear Regulatory Commission) issued the Safety Goal Policy Statement (SGPS), which included two Qualitative Safety Goals (QSGs) and two Quantitative Health Objectives (QHOs) [17], further guiding the application of PSAs. By 1992, 74 PSA reports had been completed for 106 nuclear power plants in the U.S. In 1992, some industry leaders began advocating for NRC commissioners to base regulatory enforcement on probabilistic risk assessment results [14]. By 1995, the use of probabilistic risk assessments had become well established in the nuclear industry [14]. In 1995, the Electric Power Research Institute (EPRI) released the *PSA Application Guide* to assist the industry in using probabilistic safety assessments for decision-making processes [18], during which period the concept of Living PSAs was proposed [19]. By 1999, the International Atomic Energy Agency (IAEA) had published the IAEA-TECDOC-1106 technical report, which provided the definition of Living PSAs and guidelines for their development and maintenance [11]. By 2022, Muhammad Zubair applied a fault tree model and probabilistic safety assessment methods to analyze the risk of Loss of Feedwater (LOFW) accidents in the main feedwater system of the APR-1400 nuclear power plant [20]. As of now, PSAs in most nuclear power plants worldwide have been maintained as Living PSAs, which have undergone three stages of development [21–24]:

Offline Living PSA: In this stage, nuclear power plants use offline methods to model assumed plant configurations or perform risk assessments after equipment failures. Its concepts are related to basic PSAs; therefore, at this stage, there will be no modeling of the current state of operating plants or future configuration changes.

Online and Time-Independent Living PSA: This stage features more flexible modeling and the development of specific human–machine interaction applications. It is capable of calculating the instantaneous risk associated with nuclear power plants. Changes in plant configuration must be manually and periodically input by operators. A representative example is the essential system safety monitor (ESSM).

Online Dynamic Living PSA: This stage allows for real-time monitoring and management of dynamic activities within a nuclear power plant. When the plant configuration changes, operators update the system. It is dynamic, as all redundant components are modeled based on their last inspection time. This stage's computer system is known as the Dynamic Risk Monitor (DRM).

In 2020, Chen proposed a four-stage Living PSA characterized by automatic model updating. It was a time-dependent Living PSA that combined online nuclear power plant (NPP) status monitoring and predictive technologies with automatic signal updates and online data analysis and recovery [25,26].

The development of Living PSAs is shown in Figure 1.

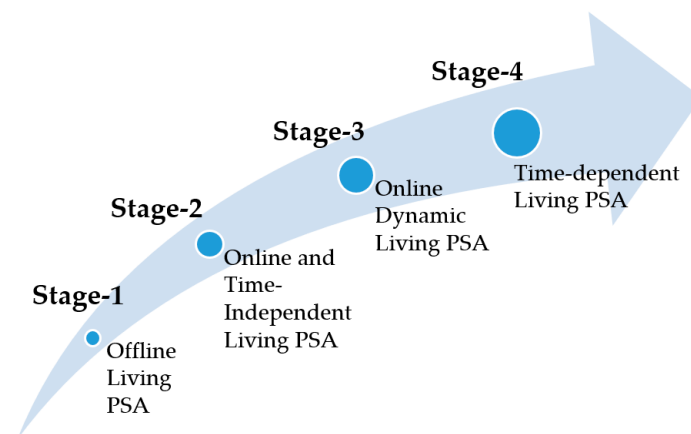


Figure 1. The development history of Living PSA.

Based on Living PSAs, nuclear power plants have introduced risk monitors to manage and monitor plant operations. A risk monitor is a tool used by nuclear power plants to assess various risks during their operation, providing technical support in the risk-informed integrated decision-making process. It consists of well-developed risk monitor software and a reliable, plant-specific operational PSA model [27]. Correspondingly, the progress and application of risk monitoring software is accelerating. The first nuclear power plant risk monitoring software, the Essential Systems Status Monitor (ESSM), was initially implemented at the Heysham and Torness nuclear power plants in 1988 [28]. Several other systems, such as the ORAM (Online Risk Assessment for Maintenance—Sentinel) SENTINEL and Safety Monitor in the U.S. [29], the SAIS (Safety Analysis and Information System) in Germany [30], the TQRM (Third Qinshan nuclear power plant risk monitor) in China [27], LIPSAS (Living PSA system) in Japan [31], and Risk Watcher, developed by Scandpower in Sweden [32], have also been adopted by multiple nuclear power plants.

1.4. Basic Principles and Methods of Living PSA

The book *System Reliability and Risk Analysis*, authored by Professor E. Frankel from MIT, provided the engineering discipline with a mathematical framework and probabilistic methods [14], enabling engineers to determine the probability of system failures given uncertainties in certain operational parameters [33]. A Living PSA is a PSA analysis technique that can be executed quickly and updated promptly to ensure the validity of its analysis results [34]. A Living PSA includes two key technologies: Living PSA modeling and Living PSA model updating. Based on the working principal of a Living PSA, it can be divided into three aspects: risk assessment, risk monitoring, and risk tracking.

The real-time risk monitor constructed using the Living PSA method can instantly collect the real-time status data and operating hours of various systems and equipment in a nuclear power plant through its digital instrumentation and condition monitoring systems, continuously monitoring the plant's risk status. The risk model and reliability data are then updated by combining the real-time status data of the monitored systems, structures, and components with corresponding risk model data, model and unit operating condition data, equipment status relationship data, and unit operating condition data. After updating, the real-time risk is analyzed by the Living PSA using its computational engine algorithm. The result data provide strong support for nuclear power plant operators, helping them formulate more accurate and efficient operation and maintenance strategies. The analysis steps of the real-time risk monitor are shown in Figure 2 [35].

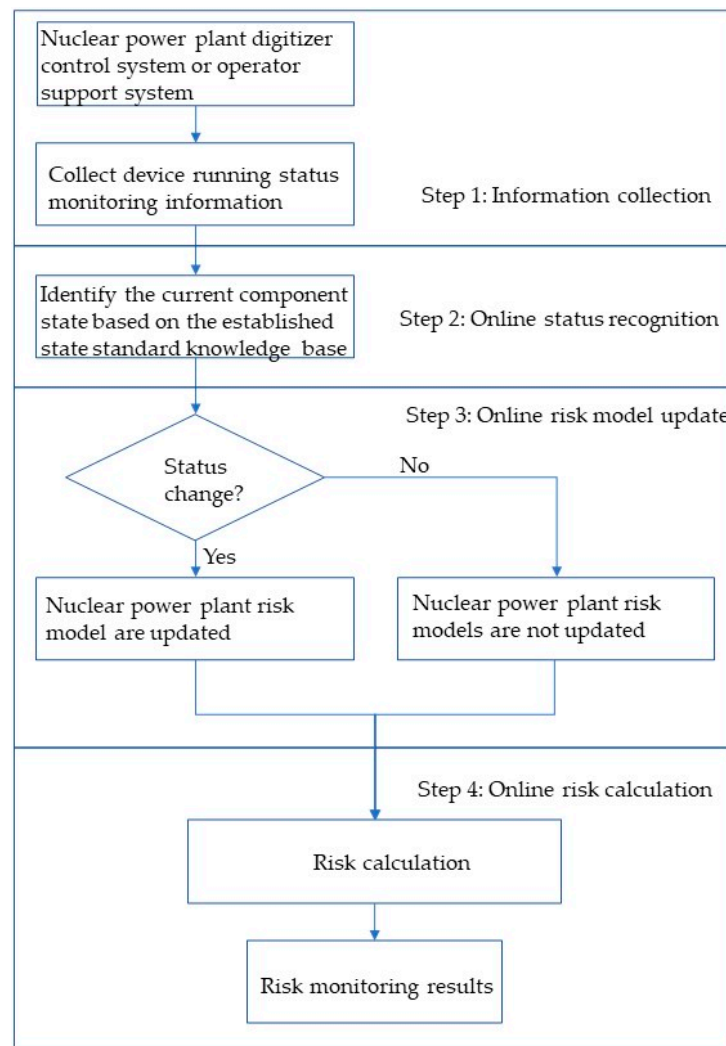


Figure 2. Analysis steps of a real-time risk monitor.

(1) Monitoring Data Collection: Real-time data of nuclear power plant equipment or systems are collected through the plant's digital instrument and control system or operator operating system. These data provide predictive conditions for equipment status detection in nuclear power plants.

(2) Online Status Identification: Nuclear power plants can construct a state space based on the actual operating conditions of PSA-related equipment. By analyzing the monitoring data to obtain the symptom space of the equipment, the current operating status of the equipment can be determined

(3) Model Updating: The online risk model in nuclear power plants typically uses a fault tree model. When the status of the equipment changes, the fault tree model must also be updated. Therefore, during the operation of the nuclear power plant, when equipment changes occur, the fault tree model must be adjusted accordingly.

(4) Risk Calculation: When the logical structure of the nuclear power plant's risk model changes, the online risk monitor will invoke the corresponding computational engine to calculate the plant's current risk metrics and present the relevant results.

2. Key Technologies of Living PSA

2.1. Modeling (Dynamic Reliability Analysis)

Within the Living PSA framework, model development is not merely about creating a theoretical or static representation; it is a dynamic and ongoing process that involves a deep understanding and precise simulation of the behavior of nuclear power plant systems and components. A well-developed model captures the complexity and variability in facility operations, providing accurate input for risk assessment. Generally, risk monitoring models are required to fulfill two key aspects: firstly, they must accurately reflect the current state of the system; secondly, they must be able to swiftly perform the analysis and calculation processes. At present, the prevailing modeling technique for a Living PSA is fault tree analysis (FTA), while, in recent years, new approaches, such as the Multilevel Flow Model (MFM) and GO-FLOW model, have emerged as well [36].

2.1.1. Fault Tree Analysis

Fault tree analysis is a well-known modeling method in PSAs for nuclear power plants [37]. The conventional FTA method was used in the renowned WASH-1400 report. FTA involves defining a system's top event, analyzing which combinations of fault conditions might trigger this top event using Boolean logic gates, and then quantifying the probabilities of these basic events to estimate the overall probability of the top event being triggered [38]. This process includes both qualitative and quantitative analyses. If the probability of the top event exceeds the acceptable level, system designers should either redesign the system or propose a more innovative design [37]. However, one limitation of traditional FTA is that it only identifies the combinations of fault components that lead to system failures but cannot analyze the specific timing of these component failures. Understanding the sequence of fault occurrences is crucial. With advancements in analysis methods, there have been many extensions to static fault trees [39], such as dynamic fault trees [40], state fault trees [41], time fault trees [42], and pandora time fault trees [43]. Another limitation of traditional fault trees is that, for quantitative analysis, corresponding failure probabilities must be assigned to the basic events in the fault tree. Traditional methods usually assume that basic events have precise probability distributions to describe their failure lifetimes. However, in practical engineering applications, this is not always the case. For example, a new complex system may not have sufficient operational experience and historical failure data to assess the reliability of its components [39]. To address the limitations in generating failure probabilities for basic events, fuzzy set theory has been used to extend standard fault trees to fuzzy fault trees [37]. By leveraging expert knowledge and experience, this approach provides a flexible method for assessing system reliability and allows uncertainty to propagate from basic events to top-level events, enabling a more comprehensive system risk assessment. In PSA applications for nuclear power plants, fuzzy probabilities are categorized into fuzzy-based FTA and fuzzy hybrid FTA [38]. Currently, the former has been applied in AP1000 nuclear power plants to mitigate large coolant leakage accidents [44], while the latter has been used in the Babcock and Wilcox reactor protection system for the David-Besse design [45]. Another practical engineering application scenario is that current risk monitoring generally uses the average failure data of a certain type of equipment to calculate the risk data, and the risk prediction ability cannot calculate the risk probability based on the actual state of the equipment unit [46]. Therefore, during the operation of the equipment, the performance changes lead to the inapplicability of the initial assignment.

Faced with this actual situation, Zhang et al. proposed a risk monitoring and prediction (PB-RMP) technology based on performance which combines predictive health management (PHM) and risk monitoring to provide a more realistic risk assessment, evaluating the changes in the importance of the equipment and predicting equipment failures and potential high-risk situations in advance. The technology was validated by a simplified Level 1 PSA model, proving the effectiveness and innovation of the method [46].

2.1.2. Multilevel Flow Modeling

Initially, the research on Multilevel Flow Modeling (MFM) originated from the challenges of representing complex systems within human–machine interfaces, particularly in the context of supervisory control scenarios. The foundational concept was conceived by Morten Lind [47]. Compared to a fault tree analysis, which analyzes systems from the perspective of failures, both the MFM and the GO-FLOW method model nuclear power plants (NPPs) from the perspective of system success [48]. MFM is a hierarchical modeling approach based on the target of a process which abstracts real physical systems from the perspectives of material flow, energy flow, and data flow; it is a graphical language used to represent the goals and intended functions of processes [49]. MFM supports goal-based problem-solving solutions and can be used for goal analysis and predicting the impact of operator behavior on system objectives [50]. The MFM approach focuses on the functional characteristics of a system, simplifying the process of system modeling and analysis and aiding in the understanding of system design and operation [48]. In terms of its philosophy and methods, it is considered a branch of artificial intelligence [47]. However, the high degree of freedom in MFM leads to significant differences in model construction among different practitioners, resulting in a substantial gap in regard to practical engineering applications. In current research, the combination of MFM and GO-FLOW methods is often employed to achieve Living PSA modeling and analyses.

2.1.3. GO-FLOW

The GO-FLOW methodology is an evolution of the GO method that was originally developed to analyze the safety and reliability of nuclear weapons and missile systems [51]. The GO-FLOW method was selected as a tool for developing Living PSAs due to its advantages in handling dynamic system configuration changes, time-dependent issues, and ease of updating and verification. Compared to traditional FTA, it offers a success-oriented approach to system reliability analysis, making it particularly suitable for assessing the availability of repairable systems with phased tasks and time-dependent characteristics [52]. This method, through its modular modeling framework, is used to analyze the dynamics and control characteristics of systems [53]. By simplifying the model updating process, it reduces manual operations and lowers the error rate. It also supports real-time risk monitoring, enhancing the timeliness of risk assessments. Additionally, the user interface design of GO-FLOW considers ease of use, allowing non-specialist users to easily perform risk management and maintenance decisions, thereby improving the safety management efficiency of nuclear power plants. The GO-FLOW method was proposed in 1988 [51]. In 2017, Dai et al. introduced a hierarchical modeling approach to enhance the efficiency of GO-FLOW in modeling, model updating, and model computation [54]. In 2016, Yang et al. proposed a dynamic PSA method for modeling various typical states of nuclear power plant components, including operation, standby, failure, maintenance, and testing. The advantage of this method is its ability to flexibly update component states based on real-time data and monitoring information without requiring structural changes to the model [55]. In 2019, Lu et al. expanded the GO-FLOW method by introducing new GO-FLOW operators to describe the control functions in terms of using the GO-FLOW model in nuclear power system risk monitoring [53]. Additionally, Lu proposed in 2021 a method for updating component uncertainty states based on human operators' cognitive processes [36]. Recently, Yang et al. proposed a comprehensive operational feasibility analysis framework based on GO-FLOW methodology which employs a pattern-matching

algorithm based on a process action library for operational monitoring and identification of potential human error patterns [56]. In 2018, Zhao et al. proposed an integrated operation monitoring and assessment system that applies MFM for goal analysis and the GO-FLOW model for reliability monitoring in advanced nuclear power plant control rooms, aiming to reduce human errors and enhance operational efficiency [50]. In 2021, Song et al. combined MFM and GO-FLOW modeling and analysis methods to propose a comprehensive decision support framework for accident emergency response planning. MFM is used to generate potential emergency response plans, while GO-FLOW is used to evaluate these plans, providing a complementary approach that offers comprehensive insights for decision-making [57].

2.1.4. Other Models

Another commonly used dynamic reliability analysis method in Living PSAs is Failure Modes and Effects Analysis (FMEA). It was first introduced in the U.S. aerospace manufacturing sector in the 1950s and was originally used to address the quality and reliability issues of military products [58]. In 2013, Muhammad et al. combined FMEA and the GO-FLOW method to conduct a quantitative dynamic reliability assessment of the passive safety systems of an AP1000 nuclear power plant [59]. In 2017, He et al. identified potential failure modes of safety valves and the specific causes behind these failure modes through FMEA. They used the FMEA results to establish a fault tree model of safety valve failures, comprehensively evaluating the overall reliability of the safety valves [60]. In 2020, Li et al. studied a Dynamic Failure Modes and Effects Analysis (DFMEA) method based on simulation technology for a reliability analysis of nuclear power plant digital distributed control systems (DCSs) [61]. In 2021, Lo et al. proposed a hybrid risk assessment model combining FMEA and the Multi-Criteria Decision-Making (MCDM) concept to improve risk assessment methods. The applicability and effectiveness of this approach were validated through a case study involving a nuclear power plant steam turbine [62].

The Markov model is a method for describing stochastic processes and is widely used in modeling system state transitions [63]. There have been several studies on the application of this model in terms of direct investment and direct investment control in nuclear power plants [64]. The Markov model was introduced by Tunc Aldemir in the late 1980s in order to conduct reliability analyses of process control systems [65]. Subsequently, Papzoglou proposed the Markov reliability analysis for dynamic systems [66]. In 2010, Jiang et al. proposed a reliability assessment method based on the Markov model to analyze and predict the transition probabilities in monitoring processes within nuclear power plant digital main control rooms [67]. In 2022, Hao et al., from the China Institute of Nuclear Power Research, conducted a study on safety-grade direct current control systems (DCSs) by constructing a reliability model based on Markov processes. They innovatively used a layered iterative approach to simulate the system's redundancy characteristics, quantitatively calculating the failure rates under detectable and undetectable fault conditions, thus providing a solid quantitative basis for maintenance planning during the actual operation of nuclear power plants [63]. Given that real-world engineering problems often have unique and complex characteristics that require a comprehensive consideration of multiple factors, Jiang et al. proposed a hybrid computational framework for risk-oriented reliability analyses in the context of Living PSAs. The research methodology involves combining Discrete-Time Dynamic Event Trees (DDETs), the Markov/Cell-to-Cell Mapping Technique (Markov/CCMT), and the GO-FLOW method to create an integrated risk solution. This hybrid computational framework effectively facilitates comprehensive risk assessment in complex digital system engineering and provides accurate modeling and rapid evaluation of the time-dependent task reliability of safety systems during accident prevention and mitigation processes [68].

The Dynamic Flowgraph Methodology (DFM), like the Multilevel Flow Model (MFM), emphasizes dynamic reasoning [69]. It is a directed graph model used for system analysis that is capable of clearly illustrating the temporal and causal relationships between states

and parameters. The DFM has been successfully applied in several studies for reliability assessment of digital systems in the nuclear industry [64]. In 2009, Karanta et al. used the DFM to assess the reliability of the digital control systems of the Olkiluoto 1 and 2 units [70]. In 2015, McNelles et al. successfully applied the DFM to assess the reliability of the FPGA-based post-accident monitoring system of the Westinghouse AP1000 nuclear power plant, revealing potential failure causes, calculating failure probabilities, and guiding improvements in hardware and logic design [71]. Given that the latest nuclear power plants all use fully digital instrumentation and control systems, in 2018, Zhou et al. used the DFM to assess the reliability of the digital automatic power control system in a nuclear power plant [72]. The DFM is a promising dynamic method, but integrating the results of DFM models into the current PSA framework remains a challenge in terms of its widespread use in Nuclear Power Plant PSAs [64,73].

Most of the models discussed above are dynamic models, which are mathematical or computational models used to describe and analyze the behavior of systems over time. Even some static models have gradually evolved into dynamic models in response to the demands of real-time monitoring, with fault trees and dynamic fault trees being typical examples. There are many methods available for building models in regard Living PSAs in addition to those we have already introduced, such as Bayesian networks [74] and Petri net models [75]. Furthermore, with the development of monitoring requirements and computational capabilities, several approaches for mixed modeling have emerged, allowing for better real-time monitoring and risk assessment.

2.2. Model Updating

For the nuclear power plant risk monitoring system, once the Living PSA model is established, the subsequent task is to update the status of each component in the Living PSA model to ensure that consistency with the actual status during operation is maintained. The method of model updating also reflects the developmental stages of Living PSAs. In the first stage, a Living PSA cannot reflect the power plant's operational status during configuration changes and requires regular updates based on data provided by the information acquisition system. In the second stage, the workers update the Living PSA at the end of inspection periods, and, in the model, the failure probability of standby components is independent of time and usage. In the third stage, the Living PSA is manually updated when there are configuration changes or other modifications at the plant [36,76]. In the fourth stage, the risk model of the Living PSA relies on the plant's online monitoring and predictive technologies. This enables the model to not only update and calculate the plant's risk with each configuration change but also perform periodic risk updates, fully utilizing the time-dependent characteristics of the model for real-time updates [77]. When changes occur in the plant's operating environment, failure phenomena, maintenance activities, or reliability data, the Living PSA model can be updated online, either automatically or manually, integrating data from operator support systems, maintenance management systems, and reliability data management systems. The updated model is re-quantified by a solver engine, and the results are intuitively displayed through a graphical user interface, assisting plant personnel in real-time risk monitoring and management optimization, thereby significantly improving operational efficiency and safety [78].

Nowadays, with the introduction of the fourth stage of Living PSAs, scholars are not only proposing new modeling methods for Living PSAs but also addressing the challenges related to updating and calculating the fourth stage model. In 2011, Zubair et al. from Harbin Engineering University conducted research on data updating methods for the Daya Bay Nuclear Power Plant [79]. They proposed a reliability data updating method (RDUM) based on real-time probabilistic safety assessments by analyzing five years of operational data for the emergency diesel generators (EDGs). In 2015, Muhammad Zubair et al. discussed how to use the alpha factor model to calculate and update the unavailability of common cause failures (CCFs) in nuclear power plants [80]. Additionally, in 2017, Zhang et al. addressed the dynamic adjustment of common cause failure risk assessment

for nuclear power plants due to changes in system configuration during operation (such as component failure, maintenance, or testing). They proposed a CCF-model-updating method based on the Alpha Factor Model (AFM). This method decomposes failure probabilities, considers coupling factors and root causes, and dynamically updates the CCF model parameters in response to system configuration changes, thereby enabling real-time and accurate risk assessment for nuclear power plants [81]. In 2019, Chen et al. proposed a time-dependent modular fault tree (MFT) structure for PSAs. By studying the mechanism of combining model automatic updates with status monitoring signals, a set of mapping rules between component status monitoring signals and model data has been developed, effectively addressing issues such as model dynamic updates, risk fluctuations arising from prolonged component operations, and data reliability, making it possible to apply the four-stage Living PSA to plant risk monitoring [26]. In 2020, Lu et al. proposed an evidence-based dynamic probabilistic safety assessment method to address the impact of uncertainty on Living PSA models. This method achieved real-time updates and improved the accuracy of nuclear power plant risk assessments. The effectiveness of this approach was validated through a case study of the component cooling water system (CCWS), demonstrating its potential for enhancing the quality of risk management decisions in practical operations [36]. In 2021, Chen et al. discussed how to address various dependencies when modeling time-dependent Living PSA models, particularly those common in redundant systems within nuclear power plants, such as state dependence, sequential failures, and CCFs. They proposed methods for modeling and updating these dependencies to ensure the accuracy and timeliness of system risk assessments [82]. This also serves as a summary and extension of their earlier work on time-dependent modular fault tree modeling and updating methods [26,83]. In 2023, Zhang et al. proposed a framework that combines PHM with risk monitoring. Within this framework, they introduced a method for updating the Living PSA model based on PHM inputs. This includes structural updates, such as changing the logical value of corresponding failure events in the model to “True”, and parameter updates for calculating equipment failure probabilities and usage time formulas, thus achieving a complementary integration of PHM and risk monitoring technologies [46].

Currently, the technology for updating Living PSA models is still in a semi-offline state [53], and there are still many challenges:

(1) The updating of equipment status data primarily relies on manual input, lacking automated real-time data streams. (2) The structural differences in Living PSA models when dealing with different parts of the nuclear power plant complicate the overall model's coupling analysis. (3) The insufficient granularity of the models leads to significant uncertainty in the analysis results.

Given the complexity and cost of online model updating, the International Atomic Energy Agency recommends an update frequency of every three years [11]. This frequency aims to ensure the real-time nature of the analysis results while considering the feasibility and cost-effectiveness of implementation.

3. Future Challenges and Development of Living PSA

At the beginning of the 21st century, the nuclear industry had finally recovered from the impacts of the Three Mile Island (TMI) and Chernobyl accidents. However, the Fukushima nuclear incident once again cast uncertainty over the future of nuclear power [84], highlighting the need for safer nuclear energy. So far, PSAs have been extensively developed across various industries, particularly nuclear power, aerospace, and aviation [85]. PSAs have accumulated substantial experience in terms of being applied to large light-water reactors [86]. As the nuclear industry shifts towards small modular reactors (SMRs) [9,87], especially sodium-cooled fast reactors (SFRs), PSA applications need to be refined to account for the specific characteristics of SMRs [88]. During the conceptual design phase [87], applying Living PSA tools in a plant environment will involve challenges such as lack of data [89], insufficient time [90], and accurately representing uncertainties due to practical constraints [91,92]. Similar to trends in other industrial applications, the

nuclear sector's instrumentation and control (I&C) is undergoing a significant shift towards digitalization [93]. It is crucial to perform reliable modeling assessments for digital systems and ensure the safety and reliability of digital I&C systems while also developing methods for validating software reliability assessments.

In response, this paper has the following suggestions for the development of Living PSAs: First, data quality and reliability are the foundation of Living PSAs. However, access to high-quality data is difficult, especially when it comes to low-volume production of nuclear power plant components and systems. Future development requires improved data collection methods, the adoption of advanced monitoring technologies and enhanced fault reporting systems, and the promotion of international data sharing. Secondly, human factor error is the main cause of nuclear power plant accidents, and its probability assessment and system impact analysis are extremely complex, requiring further research on human factor engineering, the development of effective training and operating procedures, and the improvement of automation levels to reduce human error. With technological progress, such as the development of new reactor designs and digital technology, the Living PSA method must adapt to and integrate these new technologies, use more powerful computing power and algorithms to develop more accurate and efficient models (so that new designs and new technologies can be applied in Living PSAs), and, at the same time, face emerging risks and threats, such as network security and other issues. In addition, with the application and development of Living PSAs to, on the one hand, promote international cooperation and unify standards, it is necessary to solve the differences in Living PSA methods and standards in different countries and regions; on the other hand, to meet the continuous improvement of regulatory requirements and standards, Living PSA methods need to be updated in a timely manner.

4. Summary

This paper provides a comprehensive review of the application of Living PSA technology for the purposes of safety monitoring and risk assessment in nuclear power plants. It begins with an overview of the safety requirements associated with nuclear power generation, emphasizing the critical role of Living PSAs in effective safety management. Subsequently, it discusses the key technologies of Living PSAs, including dynamic reliability analysis and model updating methods. Finally, the paper summarizes the challenges faced in the implementation of Living PSAs and outlines directions for future research and development.

Overall, Living PSA technology has demonstrated its strong potential and wide application value in the safety management of nuclear power plants. In the future, Living PSA technology will achieve more precise risk monitoring for nuclear power plants through advancements in intelligence and automation, as well as integration with big data and the Internet of Things. Global collaboration and standardization will enhance the overall level of nuclear safety management while systems will also expand to address emerging risks such as cybersecurity threats and natural disasters. Meanwhile, Living PSAs will adapt to the development of new nuclear technologies, such as small modular reactors, and enhance public trust in nuclear safety by improving transparency. Through continuous technological innovations and optimization, Living PSAs will continue to provide crucial support for the safe operation of nuclear power plants in the future, driving the nuclear energy industry towards greater efficiency and safety.

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